

S.No	Particulars	Theory		Particulars	
		Ample	Frequency	Amplitude	Time
1.	Message Signal	5	300	5	$T_m = x_2 - x_1$ $= 0.00333$ $f_m = 3482.5$
2.	Carrier	10	3500	10	$T_m = x_2 - x_1$ $= 0.000286$ $f_m = 3517.5$
3.	Phase Modulation Signal	—	—	10	$T_m = x_2 - x_1$ $= 0.000286$ $f_m = 3460$ $f_{min} = 500$ $f_{max} = 7700$

$$f_{min} = 500$$

$$x_2 = ?$$

$$x_1 = ?$$

SIMULATION OF PHASE MODULATION SIGNAL

AIM:

To generate and simulate a phase modulation (PM) signal using the Scilab and observe the message signal

Apparatus Required:

- * Laptop
- * SCI LAB Software

CODE:

clc;

clear;

close;

$A_m = 5;$

$A_c = 10;$

$f_m = 300;$

~~$f_c = 3500;$~~

$f_s = 45000;$

$k_p = 3.5;$

$t = 0:1/f_s:2/f_m;$

Calculation:

1. Message Signal:

$$\begin{aligned}T &= x_2 - x_1 \\&= (4.444 \times 10^{-3}) - (1.00 \times 10^{-3}) \\&= 3.333\end{aligned}$$

$$\begin{aligned}f &= \frac{1}{T} \\&= \frac{1}{3.333 \times 10^{-3}} \\&= 300.03 \text{ Hz}\end{aligned}$$

2. Carrier Signal:

$$\begin{aligned}T &= x_2 - x_1 \\&= (1.289 \times 10^{-3}) - (1.000 \times 10^{-3}) \\&= 2.89 \times 10^{-4} \text{ s} \\f_c &= \frac{1}{T} \\&= \frac{1}{2.89 \times 10^{-4}} \\&= 3500 \text{ Hz}\end{aligned}$$

$$m = A_m * \sin(2 * \pi * f_m * t)$$

$$c = A_c * \sin(2 * \pi * f_c * t)$$

$$p_m = A_c * \sin(2 * \pi * f_c * t + k_p * m);$$

subplot (3, 1, 1);

plot (t, m);

title ("Message Signal");

x label ("Time (s)");

y label ("Amplitude");

subplot (3, 1, 2);

plot (t, c);

title ("Carrier . signal");

x label ("Time (s)");

y label ("Amplitude");

subplot (3, 1, 3);

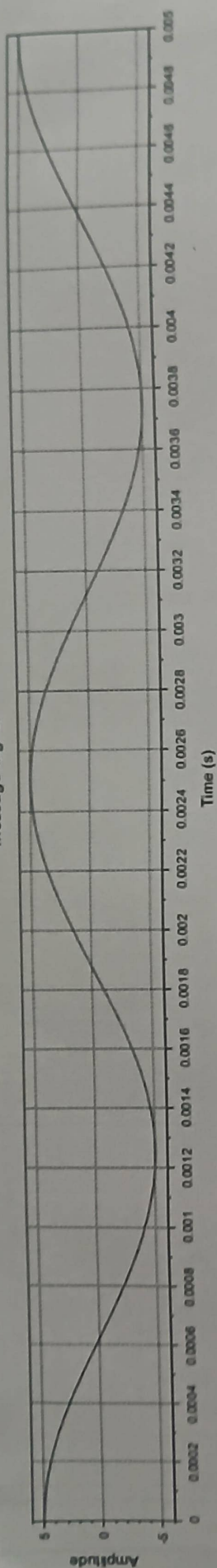
plot (t, p_m);

~~title~~ ("Phase Modulated. (PM). signal");

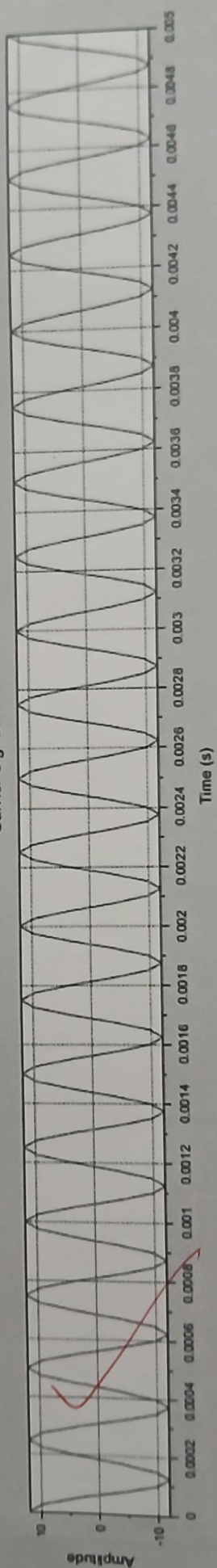
x label ("Time (s)");

y label ("Amplitude");

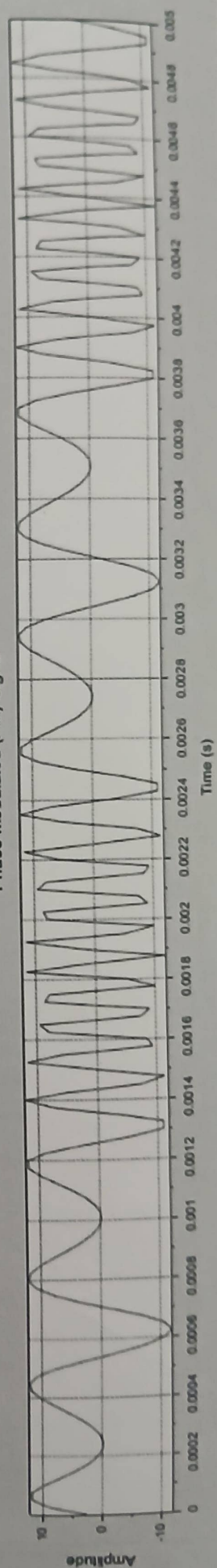
Message Signal



Carrier Signal



Phase Modulated (PM) Signal



Phase modulation:

$$T = x_2 - x_1$$

$$= 8.71 (4.289 \times 10^{-3}) - (2.000 \times 10^{-3})$$

$$= 2.89 \times 10^{-4} \text{ s}$$

$$f_p = 1/T$$

$$= \frac{1}{2.89 \times 10^{-4} \text{ s}}$$

$$= 3500 \text{ Hz}$$

$$\Delta f = A_m \times k_p \times f_m$$

$$= 5 \times 12 \times 300$$

$$= 3000$$

$$f_{\min} = f_c - \Delta f$$

$$= 3500 - 3000$$

$$= 500$$

$$f_{\max} = f_c + k_p \times f_m$$

$$= 3500 + 12 \times 300$$

$$= 7700$$

Scilab program Implementation (20)	Wave form tabulation (20)	Calculation & Result (20)	Viva Voice (20)	Record result observation (20)	Total mark
20	20	20	17	20	97

RESULT :

Thus the phase modulated (PM) signal was successfully generated and simulated using SCILAB.